

Andrew Zagula

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EDUCATION

California Institute of Technology

B.S. in Computer Science

Pasadena, CA

Expected Jun 2030

Bridgewater-Raritan High School

Salutatorian | GPA: 5.0 | SAT: 1570

Bridgewater, NJ

Sep 2022 – Jun 2026

- **Awards:** 2x USAMO Qualifier, 5x AIME Qualifier, USACO Gold, Y Combinator Startup School 2026
- **Coursework:** Calculus III, Differential Equations, Linear Algebra, AP Calculus BC, AP Statistics, AP Physics C, AP Chemistry, AP Biology, AP Macroeconomics, AP Computer Science A, Data Structures (Rutgers University)

EXPERIENCE

Alumhub

Co-Founder

Bridgewater, NJ

Jan 2025 – Mar 2026

- Co-founded student–alumni social networking platform; led team of 10+ in development and growth, scaling to 5K+ users in 250+ U.S. high schools; secured \$50K+ in platform credits from Google, Vercel, Stripe, and others
- Led full-stack Next.js development; built user-matching engine with schema-validated LLM generation and deterministic fallback ranking, real-time Firebase messaging, and Stripe billing with pro-tier feature access

University of California, Berkeley

Student Researcher

Remote

Jul 2024 – Dec 2025

- First-authored AutoAdv (NeurIPS Lock-LLM 2025), an adaptive adversarial prompting framework for black-box multi-turn LLM jailbreaking; achieved 99% attack success rate on Qwen3-235B, improving 24% over 6 turns
- Implemented two-phase prompt rewriting framework for Grok 3 Mini; developed temperature manager adjusting LLM sampling via 4 strategies, and pattern manager logging 28 effective rewriting techniques to guide future attacks
- Integrated provider-agnostic API routing and 10-worker parallel execution with per-turn tracking of token usage, latency, and cost; designed GPT-4o mini scoring function evaluating 3 success indicators for jailbreak classification

Boston University

Research Intern (RISE)

Boston, MA

Jun 2025 – Aug 2025

- Co-authored BabyVLM-V2 (CVPR 2026), a developmentally inspired framework for sample-efficient vision-language models; pretrained 1.4B BabyLLaVA-V2 on 1M+ multimodal pairs and instruction-tuned on 110K+ samples
- Constructed DevCV Toolbox, a 10-task benchmark adapted from NIH Baby Toolbox; tasks generated from 478 hours of infant-perspective (SAYCAM) video data to assess spatial reasoning, memory, and vocabulary in VLMs
- Developed OpenCV pipeline with frame sampling and sliding-window segmentation; applied YOLO for object detection and tracking, filtering 2.2K+ usable clips (~63% yield) from 3.5K+ candidates via Label Studio

PROJECTS

PaperTrail | *FastAPI, Next.js, TypeScript, LangChain, LangGraph, ChromaDB, SQLite*

- Built local arXiv research assistant with arXiv URL/PDF ingestion, LLM-generated multi-query paper discovery, structured paper breakdowns, and 3 LangGraph workflows for comparison, ideation, and implementation planning
- Developed provider-agnostic RAG pipeline using LangChain for multi-turn Q&A with section-level citations; integrated ChromaDB top-5 retrieval and SQLite storage for papers, chats, and generated research artifacts

AUDIT | *Python, OpenAI API, ChromaDB, SQLite, Typer, PyInstaller, npm*

- Built CLI security scanner to detect vulnerabilities in repositories with RAG over 50-pattern CWE/OWASP knowledge base, generating JSON reports with file/line evidence, confidence scores, and remediation guidance
- Engineered canning pipeline with high-signal candidate extraction, heuristic prefiltering, ChromaDB semantic retrieval, parallel LLM review, and content-hash SQLite caching, reducing LLM-reviewed candidates by ~70%

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, Java, SQL

Frameworks: React, Next.js, Node.js, FastAPI

Developer Tools: Git, Docker, PostgreSQL, Firebase, SQLite, ChromaDB

Libraries: PyTorch, Hugging Face, NumPy, Pandas, OpenCV, Tailwind CSS